Deriving the Work–Energy Theorem from Newton's Second Law

Starting Point: Newton's Second Law

We begin with Newton's second law in one dimension:

$$F = ma = m\frac{dv}{dt} \tag{1}$$

Our goal: show that this leads directly to

$$W = \int_{x_i}^{x_f} F\,dx = \Delta\left(\tfrac{1}{2}mv^2\right). \tag{2}$$

The Derivation

Step 1: Define work.

Work is force integrated over displacement:

$$W = \int_{x_i}^{x_f} F\,dx \tag{3}$$

Step 2: Substitute $F = m\frac{dv}{dt}$.

$$W = \int_{x_i}^{x_f} m\frac{dv}{dt}\,dx \tag{4}$$

Step 3: Apply the chain rule.

This is the key calculus move. We use the fact that velocity is $v = dx/dt$, so we can rewrite dx as $v\,dt$:

$$W = \int_{t_i}^{t_f} m\frac{dv}{dt}\,v\,dt \tag{5}$$

Here we've changed the integration variable from x to t, with the limits changing accordingly from spatial (x_i, x_f) to temporal (t_i, t_f).

Equivalently, you can think of this as a chain rule substitution directly:

$$\frac{dv}{dt}\,dx = \frac{dv}{dt}\cdot\frac{dx}{dt}\,dt = v\,dv\cdot\frac{dt}{1}\cdot\frac{1}{dt} = v\,dv \tag{6}$$

So the integral becomes:

$$\boxed{W = \int_{v_i}^{v_f} mv\,dv} \tag{7}$$

where v_i and v_f are the initial and final velocities.

Step 4: Evaluate the integral.

This is now a straightforward power-rule integral:

$$W = m\int_{v_i}^{v_f} v\,dv = m\left[\frac{v^2}{2}\right]_{v_i}^{v_f} \tag{8}$$

$$\boxed{W = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2 = \Delta K} \tag{9}$$

This is the **work–energy theorem**: the net work done on an object equals its change in kinetic energy.

The 3D Generalization

In three dimensions, Newton's second law is $\vec{F} = m\vec{a}$ and work involves a line integral:

$$W = \int_C \vec{F}\cdot d\vec{r} \tag{10}$$

Substituting $\vec{F} = m\dfrac{d\vec{v}}{dt}$ and $d\vec{r} = \vec{v}\,dt$:

$$W = \int_{t_i}^{t_f} m\frac{d\vec{v}}{dt}\cdot\vec{v}\,dt \tag{11}$$

Now use the identity:

$$\frac{d}{dt}(\vec{v}\cdot\vec{v}) = 2\,\vec{v}\cdot\frac{d\vec{v}}{dt} \quad\Longrightarrow\quad \vec{v}\cdot\frac{d\vec{v}}{dt} = \frac{1}{2}\frac{d}{dt}\big(|\vec{v}|^2\big) \tag{12}$$

So:

$$W = \int_{t_i}^{t_f} \frac{m}{2}\frac{d}{dt}\big(|\vec{v}|^2\big)\,dt = \frac{m}{2}\left[|\vec{v}|^2\right]_{t_i}^{t_f} \tag{13}$$

$$\boxed{W = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2 = \Delta K} \tag{14}$$

Same result. The work–energy theorem holds in full generality.

Summary of the Chain Rule Trick

The entire derivation hinges on one move:

$$F\,dx = m\frac{dv}{dt}\,dx = m\,v\,dv \tag{15}$$

This converts a force–displacement integral into a velocity integral, which is trivially solvable. It's the same technique you'll see when deriving the vis-viva equation in orbital mechanics, or when connecting force laws to potential energy.